FROM THE PUBLICATION

Sensor-measured, published. The kinematic figures here are the blinded Sleip sensor readouts collected during the 138-horse pragmatic prospective controlled trial published in Frontiers in Veterinary Science 2026 and reported in the DOI above.

SCIENTIFIC REPORT · OBJECTIVE GAIT · BERTONE ET AL. 2026

Blinded, sensor-based kinematic gait analysis.

Independent smartphone-video motion analysis (Sleip) corroborated clinician lameness scoring.

Background

OBJECTIVE

Clinician lameness grading (AAEP) is subjective and can vary between raters. To corroborate the clinical outcome, we ran an independent, blinded, sensor-based motion-analysis pipeline on the same horses using Sleip (Stockholm, Sweden) — a smartphone-video kinematic platform that quantifies limb asymmetry from marker-free video.

Methods

MOTION CAPTURE PIPELINE

Standardized smartphone video was recorded at Weeks 0, 2, 4 and 12 (trot ± lunge). Videos were uploaded to Sleip's cloud pipeline, quality-filtered, and analyzed by blinded raters. Two asymmetry indices were derived per limb pair: impact (early stance) and push-off (late stance). Lower asymmetry indicates a more symmetric gait.

SAMPLE & PRE-SPECIFIED ANALYSIS

The forelimb pair was the primary analysis (well-established head-nod reference). Forelimb + hindlimb combined served as an exploratory secondary analysis. Hindlimb alone was underpowered and not analyzed further.

INDEPENDENCE FROM CLINICAL SCORING

Sleip analysts had no access to treatment assignment, AAEP scores, ultrasound reads, or veterinarian notes. Video acquisition, upload, and analysis were fully decoupled from the clinical exam.

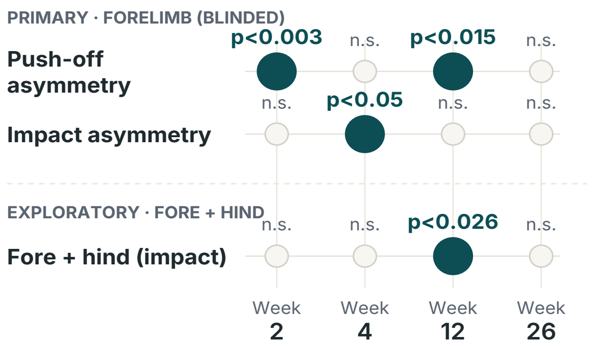
Results

FORELIMB ASYMMETRY — ECTM VS. STANDARD OF CARE

3 / 3 tests

Push-off Wk 2 $p<0.003$ · Impact Wk 4 $p<0.05$ · Push-off Wk 12 $p<0.015$

FIG 1 · Forelimb asymmetry — blinded Sleip analysis



Significant eCTM vs. standard of care comparisons across Sleip's blinded asymmetry indices. Push-off separates first at Week 2; impact reaches significance at Week 4; push-off remains significant through Week 12.

Interpretation

OBJECTIVE CORROBORATION OF CLINICAL SCORING

Because the Sleip pipeline was independent, sensor-based, and blinded to treatment assignment, the significant forelimb separations at Weeks 2, 4 and 12 constitute an objective corroboration of veterinarian AAEP scoring — not a repeat measurement of the same signal. The exploratory fore+hind combined analysis showed the same pattern with higher variance, supporting the primary result rather than adding a new claim.

Objective gait — same story as the clinician.

An independent, sensor-based pipeline picked up the same eCTM advantage in forelimb symmetry that the veterinarians reported.

Equine Performance Labs

 Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. doi: 10.3389/fvets.2026.1833933 · AVMA VCT #26005934 · IACUC ETCR-2024-0329.